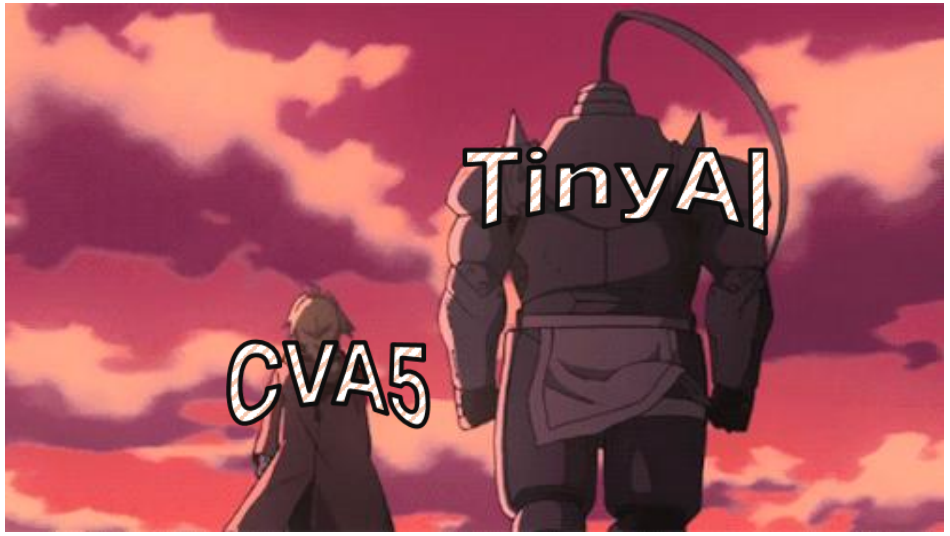


Brothers Edward & Alphonse Elric. Fullmetal Alchemist. Originally written and illustrated by Hiromu Arakawa. Anime series, Aniplex Inc.

CVA5 for TinyAI v1.0: status and RISC-V Summit North-America plans

Amirali Ebrahimi,
Alfredo Herrera,
Tao Liu,
Mathieu Gagnon,
Md Islam Tajir

GitHub: CVA5-TinyAI latest code (PYNQ-Z2)

GitHub: CVA5-TinyAI-tutorial

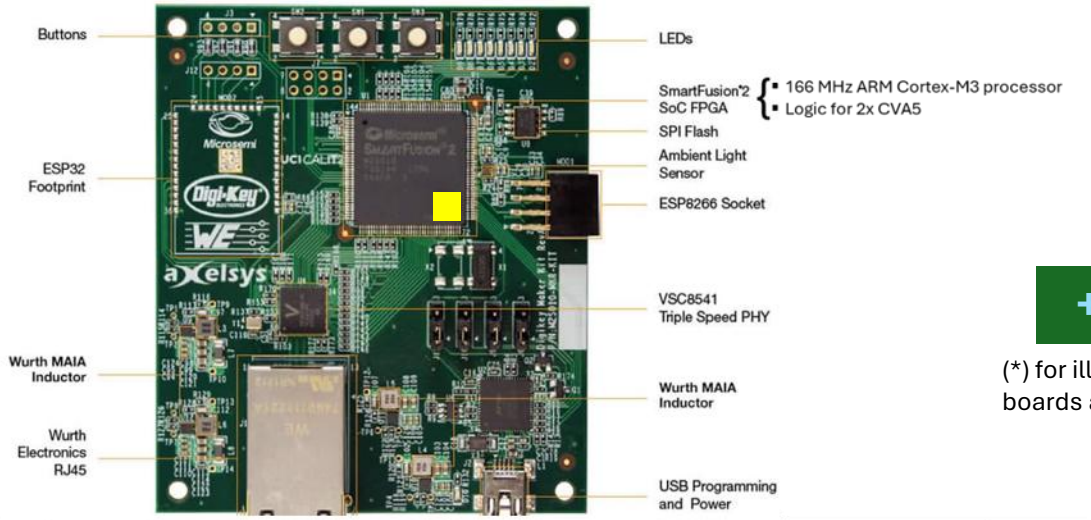
Author of F# for C# developers, developing RISC-V compiler

Status Report

PLANNED FOR VERSION 1.0	STATUS	DONE (40%)
Q3'2026: • Run verification & implementation of current CVA5 • Demo at Ottawa RISC-V user group meeting: • show CVA5 S/W running in Verilator simulation • Demo same CVA5 S/W running on FPGA • Show same CVA5 S/W running in QEMU	Oct'26 RVO meeting: • DONE • Partially done: • DONE (Jul'26) • DONE (Sep'26) • QEMU working, to demo Nov'26	95%
Q4'2026: • Develop tutorial for anyone new CVA5-TinyAI: • Systemverilog and Verilator • FPGA implementation for different eval. boards	Oct'26 RVO meeting: • DONE: OpenHW/Eclipse account • WIP: verifying CVA5 with CORE-V-VERIF • WIP: select boards (AMD, Altera, Lattice)	40%
Q1'2027: • RISC-V Summit North-America 2027	Oct'26 RVO meeting: • Presenting at RISC-V Summit N-A	25%
Q2'2027: • Complete S/W Development Kit (SDK) • Complete Develop board support package (BSP).	Oct'26 RVO meeting: • Not started yet	0%



Selecting evaluation board: FPGA ? Or micro-controller ?



+/- CAD$60

(*) for illustration only, these boards are not available anymore

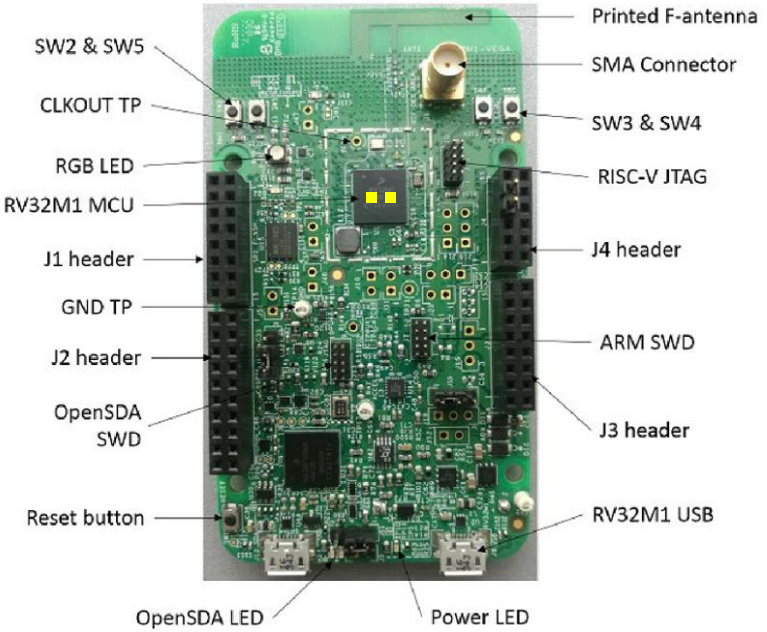
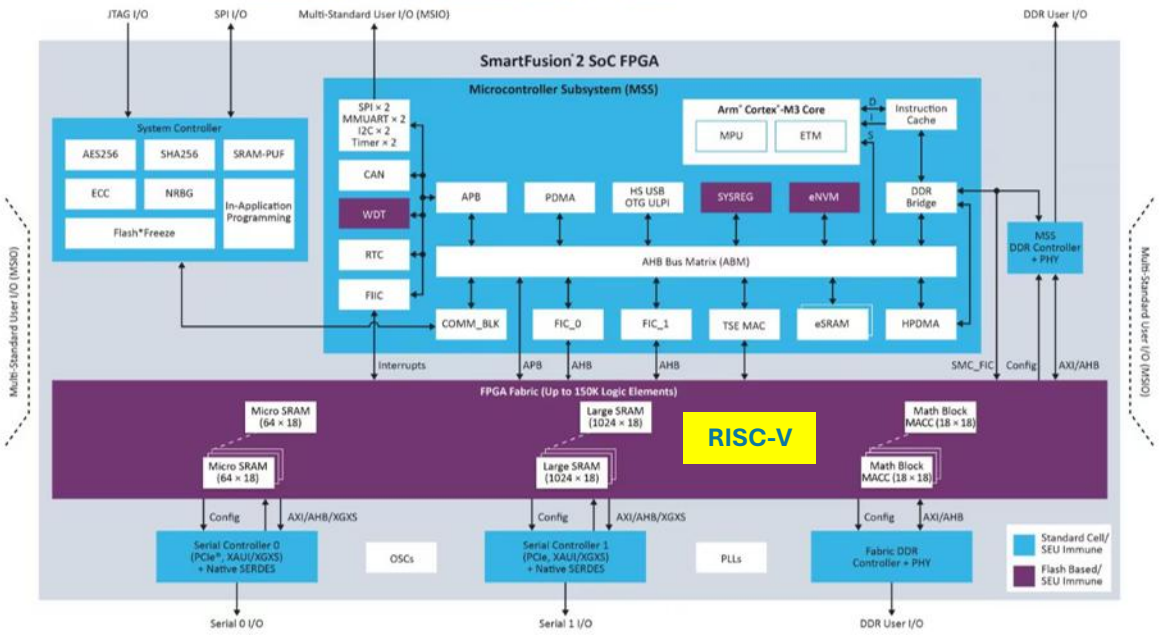
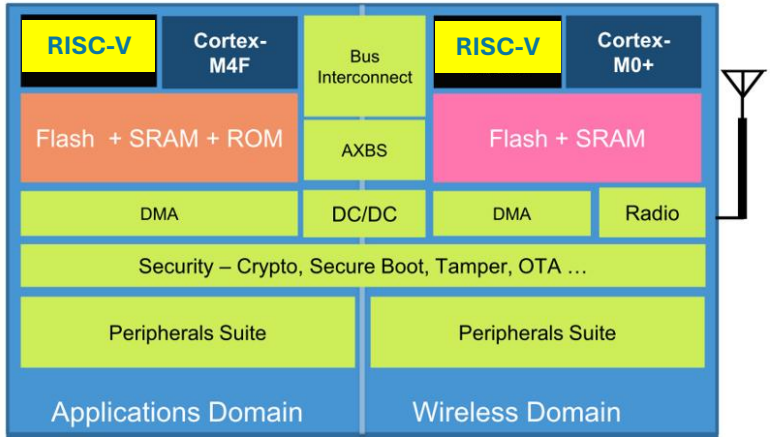


Figure 3. RV32M1-VEGA component placement





https://manuals.plus/digi-key/m2s010-mkr-kit-smartfusion2-maker-board-manual

https://github.com/open-isa-org/open-isa.org/blob/master/RV32M1_VEGA_Board_User_Guide.pdf

How does board selection impact S/W development ?

*Same compilers: standard **RISC-V GCC** or **RISC-V LLVM**.*

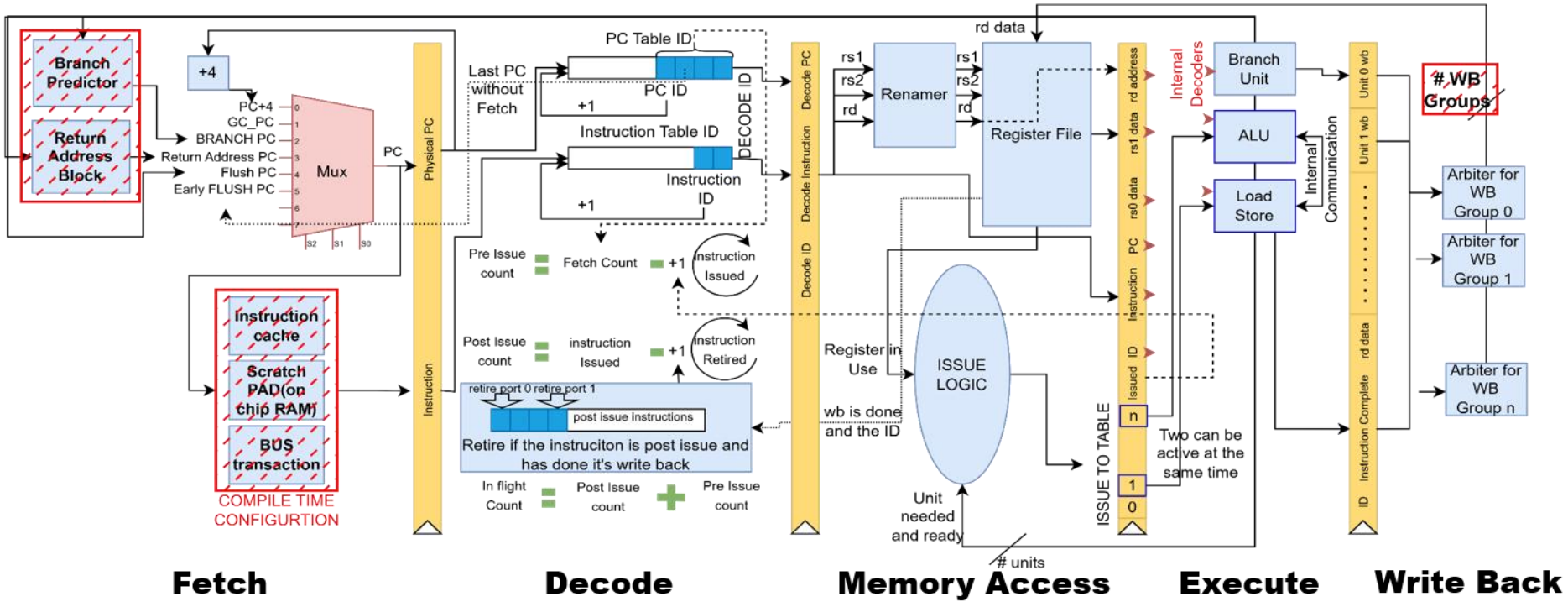
Aspect	Microcontroller	FPGA soft core	QEMU emulation virtual env.
Compiler flags	ISA fixed; the SDK sets the compiler flags	Must match synthesized core: -march=..., -mabi. Custom instructions via .insn	-cpu rv32,m=...,c=... mirrors your core; no custom instructions
Memory map & linker script	Provided by the vendor	You write it to match the HDL and reset vector	Fixed board maps: match virt
Startup code & C library [Power-up->main()] & [Std. c-lib funct.]	crt0 and newlib hooks ready	Adapt crt0; implement _write, _sbrk; fit small block RAM	Same code runs on virt's UART and RAM
Interrupts & timers	Vendor interrupt drivers in the SDK	CLINT/PLIC only if you built them	CLINT and PLIC modeled; no exact timing
Peripheral drivers	Vendor HAL / SDK	Write your own, or generate (LiteX (*), vendor BSPs)	Only modeled peripherals; custom ones need a model
Program loading	Flash over the debug probe	Baked into the bitstream, or bootloader	Load the ELF directly: seconds, no rebuild
Debugging	Probe + GDB ready	Debug module in RTL; RTL simulation	Built-in GDB server (-s -S); CI-friendly

(*) Litex commit history shows (pr#1300) added CVA5 support in 2022, contributed by CVA5's own developer, Eric Matthews. Later change requires LiteX's full installation option.

Generated and reviewed, Claude

What do you get with RISC-V source code?

- The processor unit.
- To select processor architecture, usually grouped by **Pipeline (depth) architecture**:
  - 2-stage lowRISC Ibex for small controllers
  - 5-stage CVW / Wally best for learning or teaching; spans from 32-bit embedded to full 64-bit Linux-capable processor
  - 5-stage CVA5 32-bit with floating point, specifically targeting FPGAs
  - 6-stage CVA6 64-bit production-grade, Linux-capable processor supporting ASICs or large FPGAs.
- The risk of Vendor Lock-In i.e., dependance on FPGA vendor IP blocks



Not included:

- Global clock
- Reset controller
- I/O pads/pins
- Memory
- Memory controller
- interrupt controller
- DSP blocks
- Mixed Signal blocks
- […]

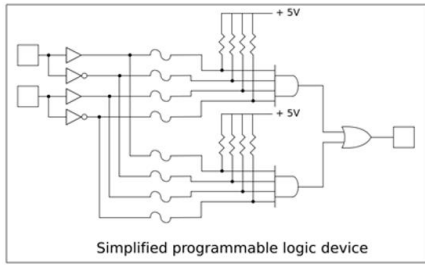
L. Shannon, C. Keilbart, and M. Shahidzadeh, "Tutorial: Incorporating CVA5, a Highly Configurable RISC-V Soft Processor, into Your FPGA Designs," IEEE FCCM 2025, May 4, 2025.

What do FPGAs offer?

- **F**ield **P**rogrammablility: inside of chip H/W functionality can be changed after device soldered on board.
- **G**ate **A**rray: grid-connected array of programmable logic that can be combined to perform complex digital functions
- **L**ook-**U**p **T**able: small memory, address-bits = logic inputs, 1-bit output

FPGA provides:

- Global clock
- Reset controller
- I/O pads/pins
- Memory
- Memory controller
- interrupt controller
- DSP blocks
- Mixed Signal blocks
- […]



Gate Array

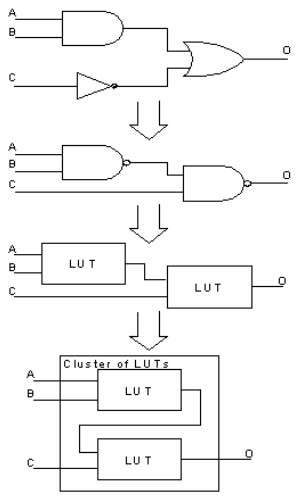
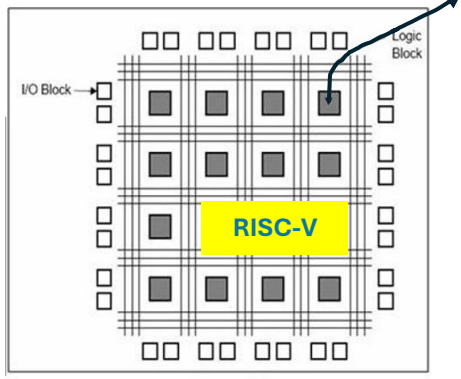
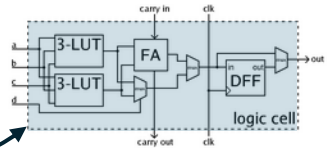


Figure 2: FPGA synthesis

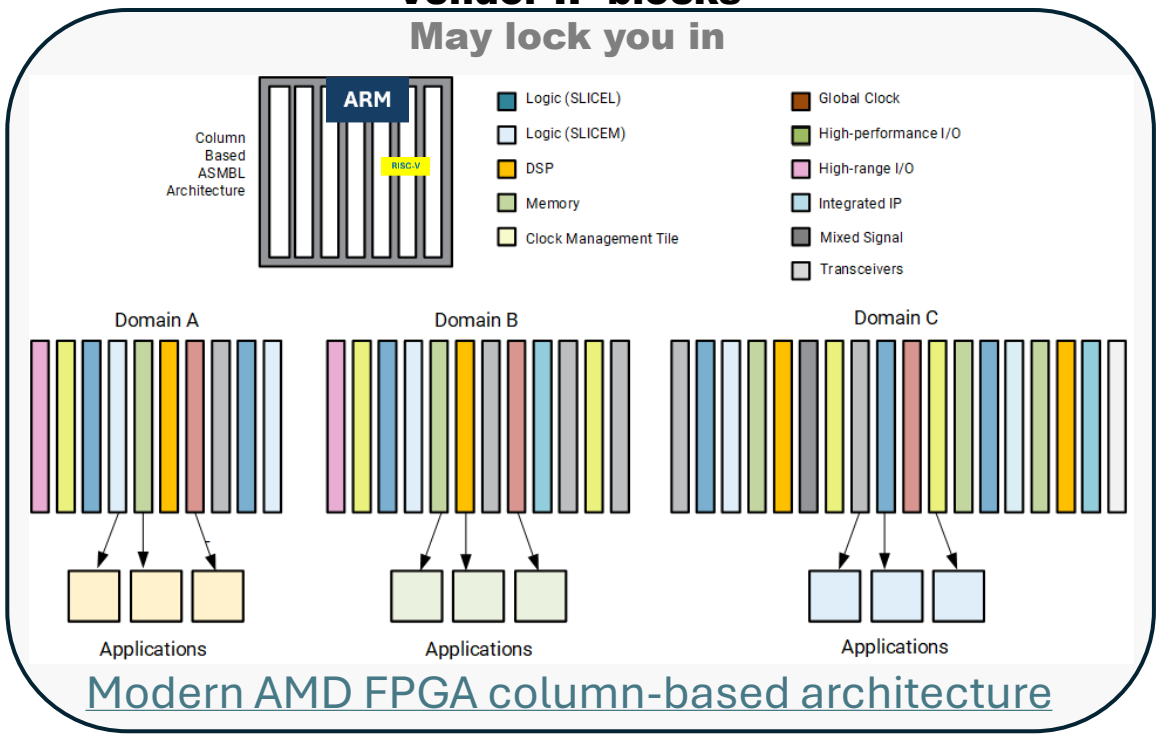


FPGA



Vendor IP blocks

May lock you in



Modern AMD FPGA column-based architecture

Y. Yao, I. Thomas, N. Kyriakopoulos, N. A. Alexandridis, and M. Younis, "Taxonomy and Evaluation of FPGA Design Tools," in *Proc. 6th Annual Military and Aerospace Programmable Logic Devices International Conference (MAPLD 2003)*, Washington, D.C., Sept. 2003.

By Petter.kallstrom – FPGA logic cell diagram, Public Domain, https://commons.wikimedia.org/w/index.php?curid=10224058

CVA5 + hls4ml CNN accelerator: porting to four boards

Code base: github.com/AmirAliEbrahimi/cva5, branch cnn-accel. Measured on PYNQ-Z2; other boards estimated.

	PYNQ-Z1 / Z2	Trenz TE0802	Lattice ECP5	Kria KV260 ★
Device · price	**Zynq-7020 · ~US$230†**	**ZU2CG · US$280**	**ECP5-85F · ~US$100–200†**	**K26 · US$249**
Logic · DSP	53.2K LUT6 · 220 DSP	47.2K LUT6 · 240 DSP	84K LUT4 · 156 mult.	~117K LUT6 · 1,248 DSP
Block RAM	140 × 36 Kb = 4.9 Mb	150 × 36 Kb = 5.3 Mb	208 × 18 Kb = 3.7 Mb	5.1 Mb + 18 Mb URAM
Hard processor	2× Cortex-A9 (unused)	2× A53 + 2× R5F	**None (CVA5 is host)**	4× A53 + 2× R5F + **GPU**
CVA5 + CNN use	**63% LUT · 52% DSP**	**~71% LUT · ~50% DSP‡**	**~60–80% LUT · 74% mult.‡**	**~29% LUT · ~10% DSP‡**
Port work	**Same part: new XDC**	Re-run hls4ml; PS clock§	**Rewrite accel + all IP**	Re-run hls4ml; PS clock§
Toolchain	Vivado/Vitis 2025.1	Vivado/Vitis 2025.1	**Radiant/Yosys; no hls4ml**	Vivado/Vitis 2025.1
Verdict	**Proven: 50.2× speedup**	**Feasible, LUT-tight**	**Major port**	**Best headroom**

The CNN: hls4ml-generated, 2 conv + 2 max-pool + 1 dense layer, 474 parameters, 14-bit fixed point; CIFAR-10 frog vs ship at 95.9%. Fed over an AXI4-Stream FIFO, it takes 305 µs per image vs 15.3 ms in CVA5 software (50.2×); 77% of that time is data movement.

Measured (PYNQ-Z2): accelerator 30,312 LUT · 115 DSP · 35 BRAM18; CVA5 3,020 LUT · 4 DSP; 64 KB local memory ≈ 15 BRAM36. **†** Price unverified. **‡** Estimated from PYNQ-Z2 numbers; ECP5 assumes 1.5–2× LUT4s. **§** PL clock and UART likely come from the Zynq PS on these boards; check board docs.

Generated and reviewed, Claude

Next steps

- Choose evaluation boards from multiple FPGA vendors:
  - Select from boards already purchased by team members… (easiest)
  - If you want to add your board to the pool, please join the project
- Prepare RISC-V Summit North-America proposal on the CVA5-TinyAI project:
  - Demonstrate CVA5-Tiny is vendor independent/agnostic
- Demo FPGA boards with CVA5-TinyAI and QEMU side-by-side
- Populate GitHub: CVA5-TinyAI-tutorial